

Institute of Actuaries of India

Subject SP8 – General Insurance: Pricing

March 2021 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:**a) Policy coverage:**

Indemnifies for claims made against Insured by clients or a third party	[1/2]
...that alleges negligence,	[1/2]
...a breach of trust	[1/2]
...or confidentiality, or defamation	[1/2]
... in their provision of services due to incorrect advice subject to policy conditions	[1/2]
This coverage can be taken by an individual or a barrister firm	[1/2]
The policy covers the cost of any damages awarded, claimants' costs	[1]
and the costs of defending the claim may or may not be covered	[1/2]
The benefit may be restricted by:	
- A maximum claim amount per event	[1/2]
- A maximum aggregate over one year	[1/2]
- An excess or deductible	[1/2]
- Insured retaining a proportion of any claim	[1/2]
- A maximum amount per claim	[1/2]
Policies are normally written on a claims-made basis	[1/2]

b) Claims Profile:

Can be long notification delays	[1/2]
Can be long settlement delays	[1/2]
Variety of different claims can be made with different profile	[1/2]
e.g. Undervaluation of Property in Divorce Proceedings may be significantly longer tailed	[1/2]
... administrative errors e.g. lost documents may be quicker to notify or settle	[1/2]
The frequency is generally not high but heavily depends upon the legislative environment	[1/2]
..., societal values	[1/2]
... and attitude towards compensation culture.	[1/2]
The class action lawsuit can make the frequency very volatile	[1]
Claim cost distributions tend to be widely spread than other class of business due to sheer nature of business and long years of court proceedings	[1]
Risk of accumulation	[1/2]
The severity / quantum of loss could be reduced due to contributory negligence exist on part of victim	[1/2]
Exposed to inflation	[1/2]
Volatility	[1/2]

c) Risk and rating factors:

The nature of the profession – type of legal advice provided by firm or individual / Area Practice	[1/2]
Size of professional firm	[1/2]
Turnover / Gross Fee Income as exposure measure	[1/2]
Experience of firm or Individual	[1/2]
Territories e.g. whether there is US exposure	[1/2]
Claims History	[1/2]
Deductible / Discount for retained interest	[1/2]
Underwriter risk assessment	[1/2]
Policy Limit	[1/2]

(Marks available 20)

[Max. 10 Marks]**Solution 2:****i) Properties of XL Scales:**

- The curve is non increasing. XL scales are denoted by $E(x) - LEV(x)$. Since, $LEV(x)$ is an increasing function, XL scales are non-increasing function.

- The curve is convex, as LEV (x) increases at a decreasing rate.

(1 mark for each point)

[Max 2]

ii) Analysis to ascertain whether the rating factors are correlated:

- The purpose here will be to find out whether the rating factors viz., Age and Driving experience, Age & Credit rating, Age & Marital Status, Driving experience & Credit rating are correlated with each other, as there is possibility that these rating factors are correlated – For example, higher the age, more the experience etc., (1/2)
- One way to check whether the rating factors are correlated in GLM is using offset. (1/2)
- One of the rating factors mentioned may be removed and the observed values may be regressed on the remaining rating factors (which is a restricted model). (1/2)
- Based on the GLM fitted on the offset linear predictor, the parameters shall be derived. (1/2)
- If the deviances between the observed and fitted values are truly random, it may be concluded that the removed rating factor is redundant and hence the rating factors are correlated. (1/2)
- Chisquared statistics, AIC, F statistics may also be used to study if the deviances are significant (1/2)
- The graph of number of claims versus ratio of (response variable derived from restricted model to response variable derived from un restricted model) may be used. The narrower, the graph is, with more values concentrated towards 1, there is more chance that restricted model is better and hence, the rating factors are correlated. (1)
- Extrinsic aliasing may also be used to identify any correlation (1/2)
- If the levels of the rating factor are perfectly correlated with each other, instead of inherent correlation, extrinsic aliasing may be used. (1/2)

(For any other reasonable analysis, ½ mark may be given for each point)

[Max 4]

iii) Constructing ILF curves:

- The claims data shall be cleaned and the older data, if any shall be brought on line with the current data and adjusted for inflation, trends etc., (1/2)
- If different excesses are different for data from different insurers, the claims shall be made ground up (1/2)
- It shall be ensured that the basic assumptions are met – Frequency is independent of limit, Distribution of claims is independent of the number of claims and the limit (1/2)
- The data shall be suitable for purpose – with / without expenses, for per person limit, per event limit etc., (1/2)
- LEV at the base level (Rs 15 lakhs) shall be first estimated. Where X is the claims amount, $E(X / X < 15,00,000)$ shall be estimated. (1/2)
- Similarly, LEV at the other sum insured (Rs 20,00,000 ... Rs 1 crores) shall be calculated. (1/2)
- ILF for base level shall be taken as 1. Then ILF at higher sum insured Rs 20 lakhs,... Rs 1 crores shall be calculated as $LEV(\text{at higher sum insured}) / LEV(\text{at base sum insured- Rs 15 lakhs})$. (1)

[Max 4]**[10 Marks]**

Solution 3:

- Price inflation: The general level of inflation in economy
 - e.g. CPI, WPI, RPI,
 - auto-parts inflation can impact the repair cost,

- rebuilding cost for household and replacement cost for content insurance policies.
- The claims for motor own damage part are also subject to these inflations along with wage inflation as some part of the overall claim will also affect by change in labor cost i.e. wage inflation.

[1/2 mark each]

- Earnings inflation: The general increase in the wage levels within the economy
 - can impact the claim amounts differently for different products as wages generally increase disproportionately for different profession.
 - Almost, all lines of business are impacted by some form of earnings inflations.

[1/2 mark each]

- Medical inflation: The general increase in overall medical expenses
 - ...inclusive of change in doctor fees,
 - medical test cost,
 - drugs and medicine cost
 - and room charges cost
 - ...impacts the health indemnity policies claims, motor third party claims
 - The medical inflation is difficult to estimate accurately in advance due to medical advancement and in absence of government regulatory body to have a check on it.

[1/2 mark each]

- Court inflation: Any trends in the magnitude of awards made by the courts will be reflected in the average severity of claims under many or few headings of claim component.
 - The court decisions can significantly increase the frequency too for admissible and payable claims
 - The liability claims which are generally decided by the courts are exposed to this type of claim inflation.

[1/2 mark each]

Main underlying drivers of super-imposed inflation:

- i.) Legislative and judicial developments
- ii.) Aggressive strategies of the plaintiffs' bar
- iii.) Shifts in judge and jury attitudes
- iv.) Evolving litigation finance

[1/2 mark each]

Impact of super-imposed inflation:

- This inflationary trend dangerously consume capital exponentially.
- While economic inflation is generally obvious, super imposed inflation can go unrecognized for many years.
- Because of these inflation trends, there could be chronic and massive under-reserving
- ...and underpricing, and this could lead to collapse or insolvency of an insurer.

[1/2 mark each]

[8 Marks]

Solution 4:

- i) (Any example may be used to illustrate the difference. A sample is given below).
 - Let us consider 2 rating factors in a data set that are correlated, where both have an influence on the response – Age of the driver and Model of the car
 - One way analysis is done by ignoring the relationship between these two factors. Two way analysis is done by considering the relationship between the two factors.
 - Let the risk relativities be as below:

	Young	Old
Age	3	1
	Low	High
Car Group	1	2

- Let us consider the exposure and Predicted value be as follows (one way analysis):

Age	Car Group	Exposure (1)	Predicted Relativity (2)	Response (3=(1)*(2))
Young	Low	10000	$3*1=3$	30000
Young	High	100	$3*2=6$	600
Old	Low	4000	$1*1=1$	4000
Old	High	6000	$1*2=2$	12000
Age	Car Group	Exposure (1)	Response (2)	Relativity (2)/(1)
Young		10100	30600	3
Old		10000	16000	1.6
	Low	14000	34000	2.4
	High	6100	12600	2.1

- Supposing the interaction factors are as below (in two way analysis):

		Low	High	
Young		1	4	
Old		1	1	
Age	Car Group	Exposure (1)	Predicted Relativity (2)	Response(3=(1)*(2))
Young	Low	10000	3*1*1=3	30000
Young	High	100	3*2*4=24	2400
Old	Low	4000	1*1*1=1	4000
Old	High	6000	1*2*1=2	12000
Age	Car Group	Exposure (1)	Response (2)	Relativity (3=(2)/(1))
Young		10100	32400	3.2
Old		10000	16000	1.6
	Low	14000	34000	2.4
	High	6100	14400	2.4

- Difference in relativity is due to interaction term.

[Max 3]

ii)

Policy Data:

- Policy Start date and end date
- Type of vehicle
- Vehicle age
- Vehicle color
- Vehicle number
- IDV
- RTO office detail
- Vehicle use location
- Driver age
- Driver experience
- Driver gender
- Excess details
- vehicle model
- vehicle make
- NCB details
- OD premium
- TP premium
- Add-on premium and other details
- Annual Mileage
- Traffic violation details
- Other relevant details for GLM modeling

Claims data:

- Policy number
- Claim occurrence date
- claim reported date
- claim payment date
- claim status
- Accident location
- Type of claim
- Claimed amount
- Payment amount
- Petition date
- age of claimant
- annual income of claimant
- type of injury
- details of injury
- Garage details
- No of dependants

[0.5 marks for each correct field]

[8]

iii)

- Since the management is not in favour of using distribution channels, the same may be done through “offset” in GLM.

(1/2)

- The parameter for this rating factor may be fixed at the level desired, in discussion with the underwriters, by analysing the claims data available with the insurer and hence the same shall not be allowed to change. (1)
- Hence the offset is subtracted from the linear predictor and the response variable (which is frequency or severity, as may be the case) is regressed on the remaining variables. (1)
- When this parameter is fixed, the other parameters tend to get readjusted, so that model explains the difference. (1/2)

[Max 3]**iv)**

- Cook's Distance, $c_i^P = \frac{h_{ii} * (r_i^P)^2}{(1 - h_{ii}) \sum h_{ii}}$ (1)

$$r_i^P = \frac{Y_i - E(Y_i)}{SD(Y_i) * \text{Root of } (1 - h_{ii})} \quad (1)$$

Hence $r_i^P = (25 - 17.2) / (\text{Root of } 15000 * \text{Root of } (1 - 0.5)) = 0.09$

(1/2)

Cook's distance = $0.5 * (0.09^2) / (1 - 0.5) * \text{Root}(1) = 0.0081$ (1/2)

Since cook's distance is much lesser than one, the deviance is not significant and the data point does not need further investigation. (1)

[4]**[18 Marks]****Solution 5:**

Individual risk model: Under the individual risk model the total claim amount S payable during a specified period in respect of a block of policies is: $S = X_1 + X_2 + \dots + X_n$

where X_i is the claim amount payable during the period in respect of risk i and n is the number of risks. [1.5]

Assumptions:

- Risks are independent
- Claim amount for risks need not be IID
- Number of risks does not change over time
- Number of claims from each risk is either 0 or 1, with probability of claim p_i , which may vary for each risk [1.5]

Individual risk model can be used for modeling the total claim amount in a group term life account. [1]

Collective risk model: Under the collective risk model the total claim amount S payable during a specified period in respect of a block of policies is: $S = X_1 + X_2 + \dots + X_N$

where X_i is the claim amount payable during the period in respect of the i th claim and N is the (random) number of claims during the period. [1.5]

Assumptions:

a) Claim Amount X_i are IID

b) The X_i s and N are independent [1.5]

Collective risk model can be used for modelling total claim amount from a motor fleet. [1]

[8 Marks]

Solution 6:

i) Silent cyber refers to potential cyber exposures contained within traditional property and liability insurance policies which may not implicitly include or exclude cyber risk.

Personal property, such as the computers damaged by some malware attack, is one example of property that can be impacted by silent cyber.

In case a hacker steals a professional's client money, could be a case of silent cyber in professional indemnity covers.

[2]

ii)

1. The inherent uncertainty in available cyber models

....and the lack of reliable claims data.

2. very difficult to model non-affirmative cyber risk

3. It may include each policy, treaty wording reviews

4. It may subject to court interpretation of wordings.

5. Accumulation risk - Risk accumulation for cyber as a line of business is already an issue for insurers and reinsurers. However, it is potentially dwarfed by that of cyber as a peril across multiple lines.

6. Continuous advancement in digital field and evolving risk.

7. Cyber attack can be or can not come under war exclusion.

8. Sometimes, significant delay in establishing the loss and quantum of losses.

[0.5 marks for each correct reason, Total 4]

[6 Marks]

Solution 7:

i)

- Legal Liability to third party
- Physical damage to Drone body / hull
- PA cover to operator
- Medical expenses cover to operator

[1/2 mark each, Max 2]

ii) **Pricing considerations and Challenges in pricing the cover:**

- Lack of data is the major concern. (1/2)
- As the cover is totally new, reinsurers' support and market data is not available. (1/2)

- Till date, the risk has not been covered before. There is not much data available for different types of covers. (1/2)
- There may be data available in other jurisdiction. Since the legal environment in the country and the litigious nature of the policyholders are different, it is difficult to use the same in pricing. (1)
- Liability amounts are not known and more so, when the third party is involved. Apart from the frequency of the cover, if the claim amount under liability cover is not capped, it is difficult to assess the severity (1)
- Even the legal environment for the cover is not known yet. As the claims under the product have not yet been taken to the court, it is not clear how the court will decide the claim amount. (1/2)
- Terms and conditions under the insurance cover is not so clear cut. It poses more difficulty in pricing. (1/2)
- The inflation and other trends is difficult to be determined. As the cover is not comparable with marine etc., it poses issues in pricing. (1/2)
- The expenses incurred are difficult to be ascertained, as specialised skill may be required in underwriting. (1/2)
- Still the regulatory environment for drones is not clear – whether it will be made mandatory for all drone operators etc., As this determines the behaviour of policyholders, this plays an important role in pricing. (1/2)
- The following plays an important role in pricing : Excess level, Pilot experience, Type of use, Geography of usage etc., There is not much study around these areas. (1)
- There may also be damage to nearby other flying objects – for example, flights etc., The liability under the same yet to be studied or whether the same will be covered under the insurance policies of the aircrafts etc., (1/2)
- When the policy is covered on “Pay as you go” basis, it poses further problems as the anti-selection risk increases under the same. Same shall also be considered in pricing (1/2)

[Max 8]

iii) Addressing problems stated above:

- Reinsurers’ support may be taken based on experience in other markets viz., UK, US, Australia etc., where the insurance covers have been introduced. The same may be modified based on difference in coverage, deductible, liability capping etc. (1)
- Similar covers if any available with the insurer, For example, aviation insurance (hull, liability etc.,) may be considered while pricing the cover. It may be more helpful for considering the liability cover (1)
- Recent claim data in either the home market or other market collected through other various sources viz., industry data, if any available could be used. (1)
- Liability shall be capped, instead of offering unlimited liability for the policy. Only benefit based policy could be offered, which aids in fixing the severity under the product (1)
- Frequency of claims may be estimated based on type of use, geography if available based on other data sources (1/2)
- The underwriters and claims personnel support may be taken if expertise is available (1/2)

[Max 5]

[15 Marks]

Solution 8:

i) Assumptions in Bulmann Starub model:

The Bühlmann-Straub assumptions are that there exists a latent parameter θ_i (which can be characterised as a risk profile and which in itself is a random variable) such that:

$$E(X_{ik} | \theta_i) = \mu(\theta_i) \quad (\text{that is, the underlying long-run claims ratio of risk } i)$$

$$\text{var}(X_{ik} | \theta_i) = \frac{\sigma^2(\theta_i)}{V_{ik}} \quad (\text{that is, the variance of the observed claims ratio of risk } i \text{ is inversely proportional to the volume measure of risk } i).$$

Other assumptions are:

- The i th risk is described by the pair, $(\theta_i, (X_{ik})_{k \geq 1})$ where $(X_{ik})_{k \geq 1}$ is the sequence of claims ratios observed for risk i in years k .
- The pairs $(\theta_i, (X_{ik})_{k \geq 1})$ are mutually independent.
- The θ_i are independent and identically distributed.
- Conditionally on θ_i , the X_{ik} 's are independent (but not necessarily identically distributed).

[1/2 mark for each point]

[3]

ii) $P(\text{Claims in Northern part}) = \frac{3}{4}$

$$P(\text{Claims in Southern part}) = \frac{1}{4}$$

$$E(\text{Claim count} / \text{North}) = 1 * (1000 / (9000 + 1000)) = 0.1 \quad (1/2)$$

$$E(\text{Claim count} / \text{South}) = 1 * (810 / (810 + 8190)) = 0.09 \quad (1/2)$$

$$E(\text{Claim} / \text{North}) = 10000 * 0.2 + 20000 * 0.3 + 30000 * 0.5 = 21200 \quad (1/2)$$

$$E(\text{Claim} / \text{South}) = 35000 * 0.7 + 45000 * 0.3 = 38000 \quad (1/2)$$

$$\mu(\Theta_A) = 0.1 * 21200 = 2120 \quad (1/2)$$

$$\mu(\Theta_B) = 0.09 * 38000 = 3420 \quad (1/2)$$

$$\mu = 2120 * \frac{3}{4} + 3420 * \frac{1}{4} = 2445 \quad (1/2)$$

$$\lambda = 2120^2 * \frac{3}{4} + 3420^2 * \frac{1}{4} - (2445)^2 = 316875 \quad (1/2)$$

$$\Phi = 6500000 - 316875 = 6183125 \quad (1/2)$$

$$Z1 = 1 / (1 + 6183125 / 316875) = 4.88\% \quad (1/2)$$

$$X1 = 25000000 / 10000 = 2500 \quad (1/2)$$

$$\text{Bühlmann Straub premium} = 4.88\% * 2500 + (1 - 0.04875) * 2445 = \text{Rs } 2448. \quad (1/2)$$

Other assumptions made are:

Here the number of policies has been taken equal to the number of claims, so that premium per exposure has been derived. (1)

[Max 7]

[10 Marks]

Solution 9:

i) Let X be the loss ratio; Rs 100 be the premium

$$100-20-15-15+(0.15+0.05)*20 - 100X+ 0.2 * 100X+0.05 * (100-20-15-15) - 0.05/2* 100X = 3 \quad (1.5)$$

Here, the investment income on premium less commission less expenses less ceded premium has been taken for 1 year; and investment income on claims has been deducted for half year. (1)

$$56.5 - 82.5X = 3$$

$$X \text{ is } 64.8\% \quad (0.5)$$

[Max 3]

ii) Events in Cat model for floods:

- Flood similar to Chennai floods in 2016 etc.,
- Flooding with 85% of area of the location cut off due to flood.
- Flood with flood magnitude value of more than 6.
- Flood with intensity which happens once in 200 years.

[1/2 mark each]

[Max 2]

[5 Marks]

Solution 10:

(i) Below business angles can be looked into:

- Overall distribution network to achieve economies of scale.
- combined UW and claim management units and related expense reductions
- any diversification benefits

(ii.) Analysis through various accounting ratios, P&L statements, income statement and balance sheets

—

- Overall investment returns and quality of asset and its mix
- Profit margin and return on capital employed
- Loss ratio, combined ratio and operating expense ratio
- Various liabilities – tax, contingent and any other liability

(iii.) The adequacy of various reserves should be estimated. Also investigate reserving trends in UPR and IBNR by comparing the reserves of various years to identify trend, if any.

- frequency and average cost and trend in claims
- analyse the above by different cohorts of claim years (e.g. AY, DY and PY)
- Being into fire Lob special attention to check for any large losses
- Coinsurance details
- Existing arrangement of reinsurance treaties and facultative arrangements

(iv.) Look into the trend in premium rates to get idea of strategical direction of business.

- market share and trend in the market share
- new business levels and trends

(v.) Attentions should be given to statutory solvency levels, free surplus, credit rating of the company and various assets mix supporting different liabilities.

[10 marks]
